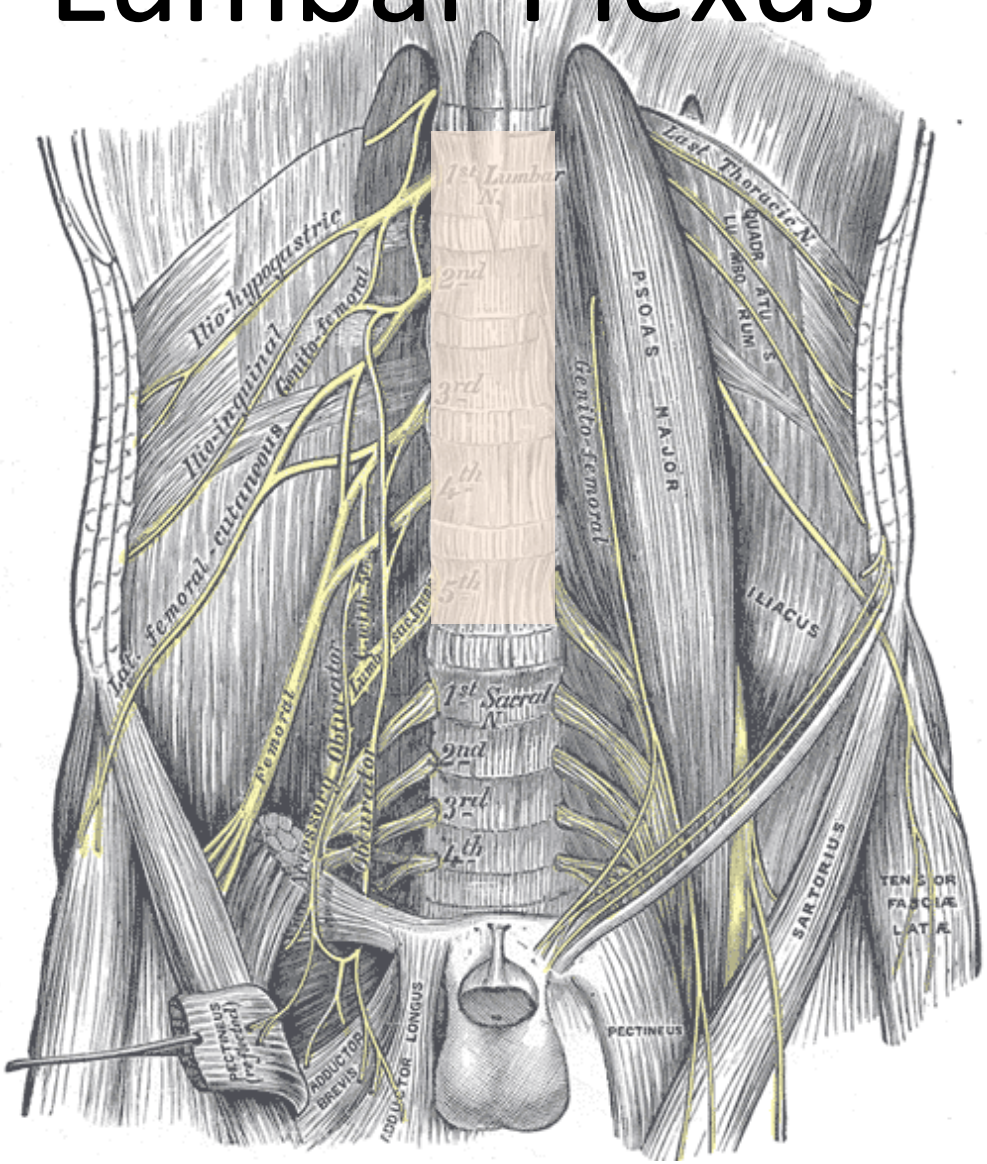
Lumbar Plexus

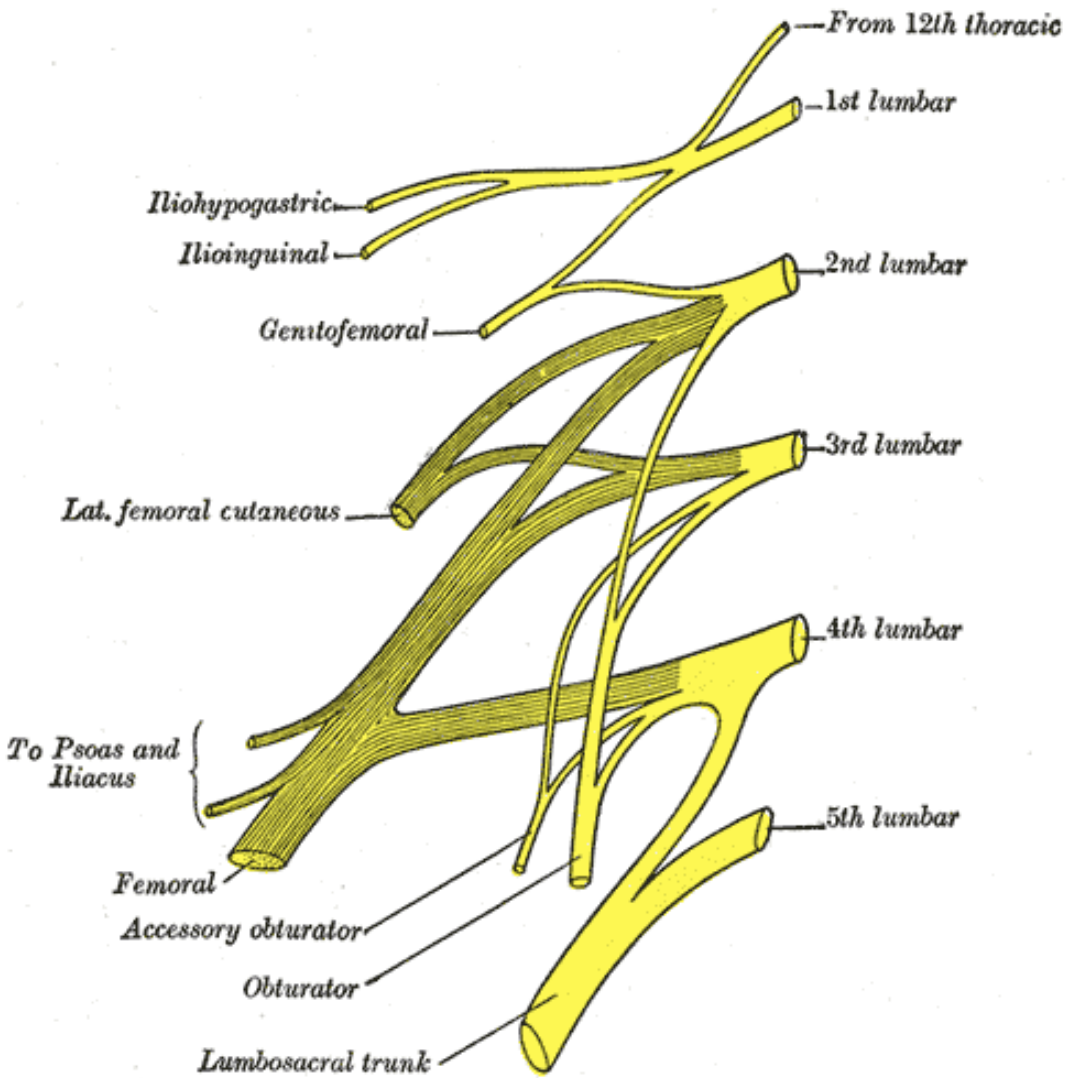


Learning Objectives

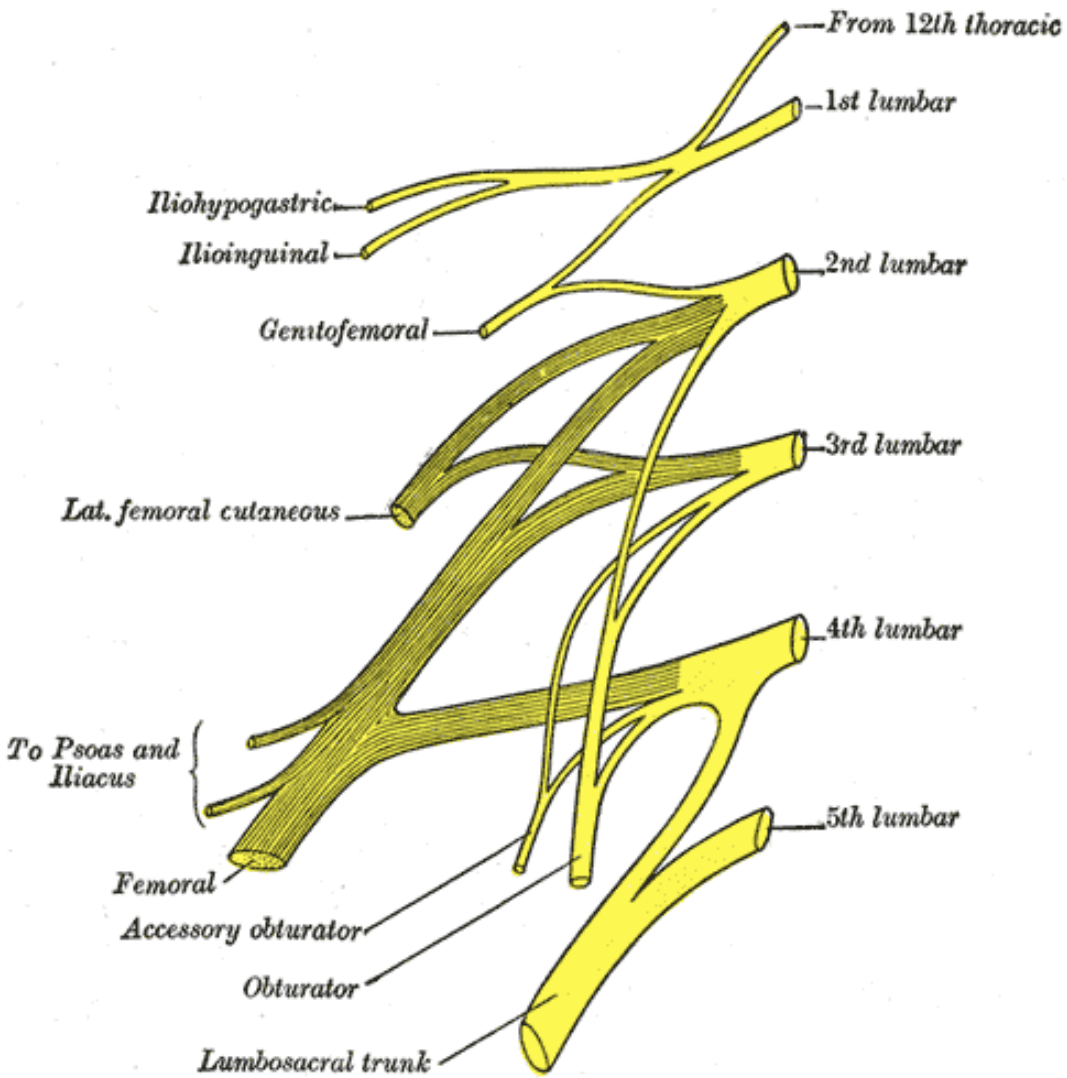
Enumerate the six key nerves associated with the lumbar plexus
Appreciate their origin, course and distribution
Be familiar with how nerve integrity can be tested clinically

- Iliohypogastric nerve
- Ilioinguinal nerve
- Lateral cutaneous nerve
- Genitofemoral nerve
- Femoral nerve
- Obturator nerve

The Lumbar plexus



The Lumbar plexus

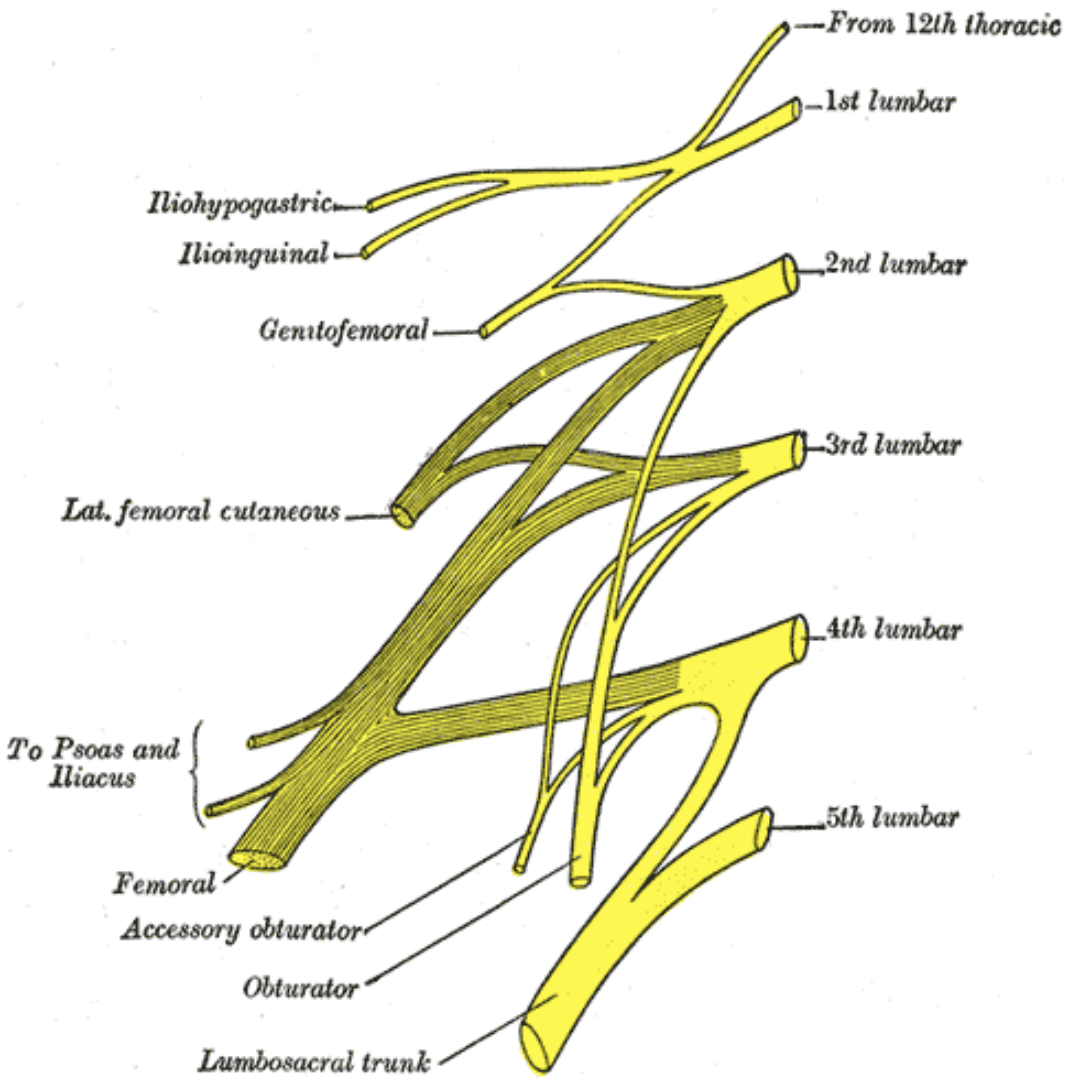


2 nerves from 1 root

2 nerves from 2 roots

2 nerves from 3 roots

The Lumbar plexus



2 nerves from 1 root

Iliohypogastric n.
Ilioinguinal n.

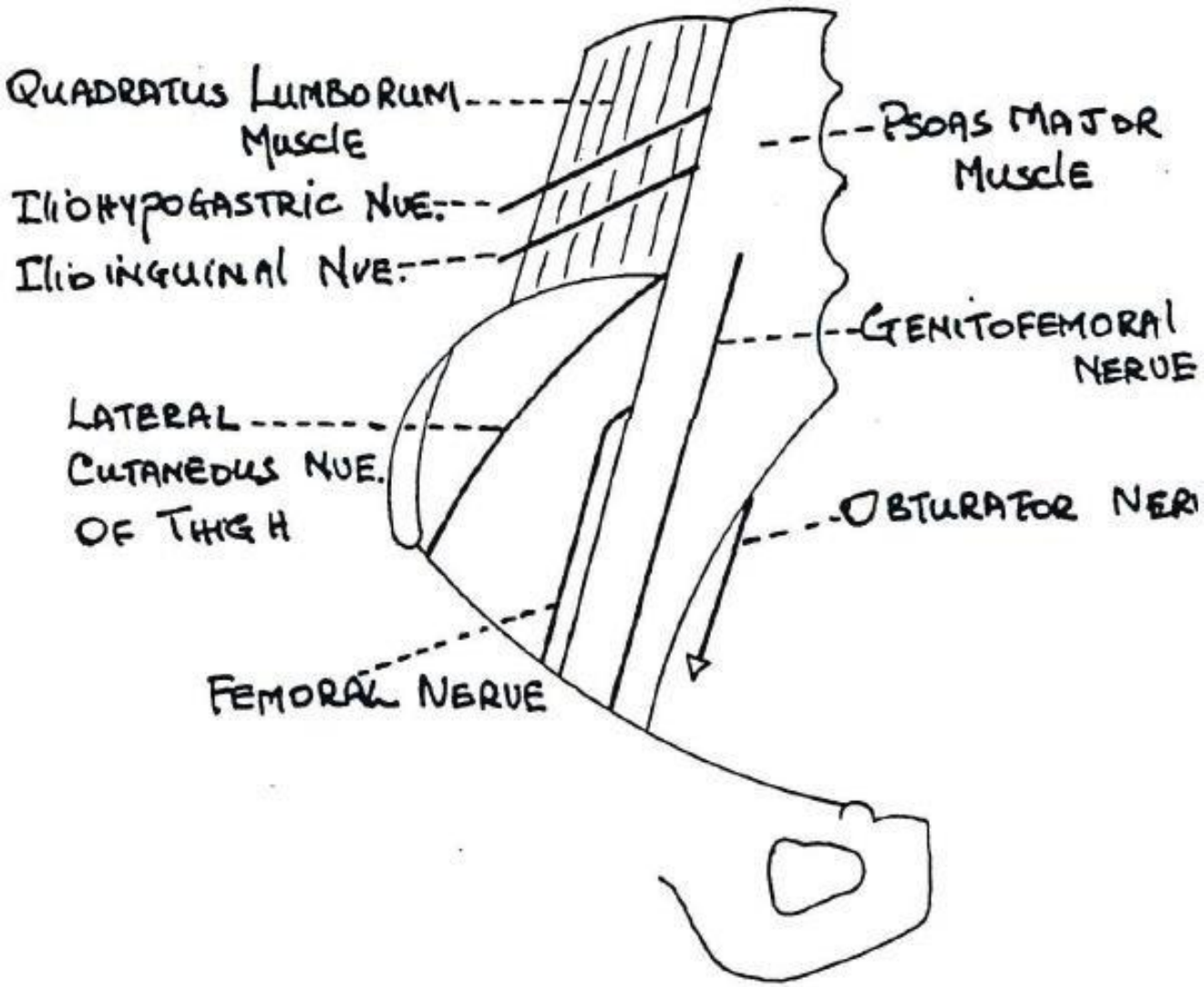
2 nerves from 2 roots

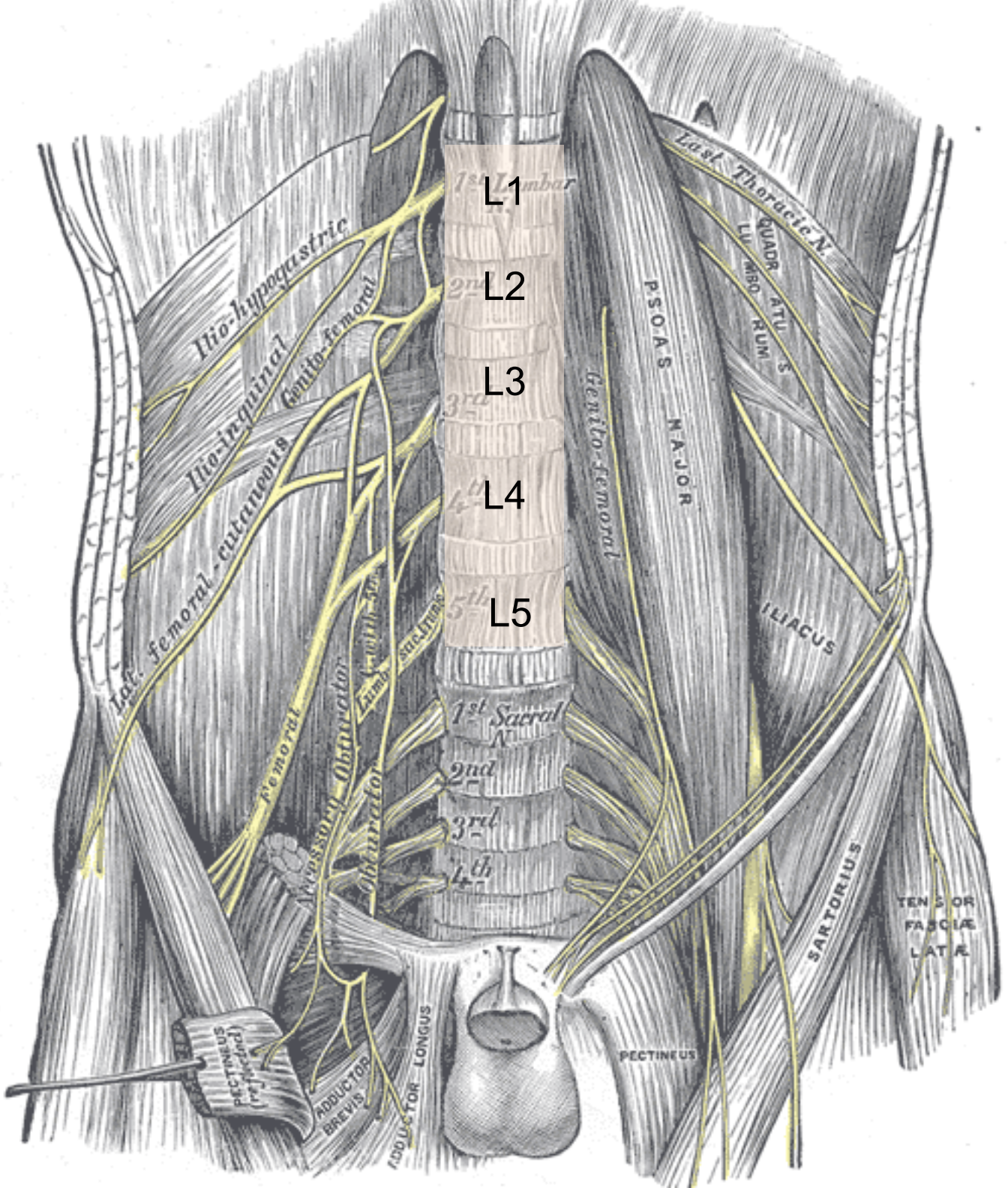
Genitofemoral n.
Lat. Cut. N. of thigh

2 nerves from 3 roots

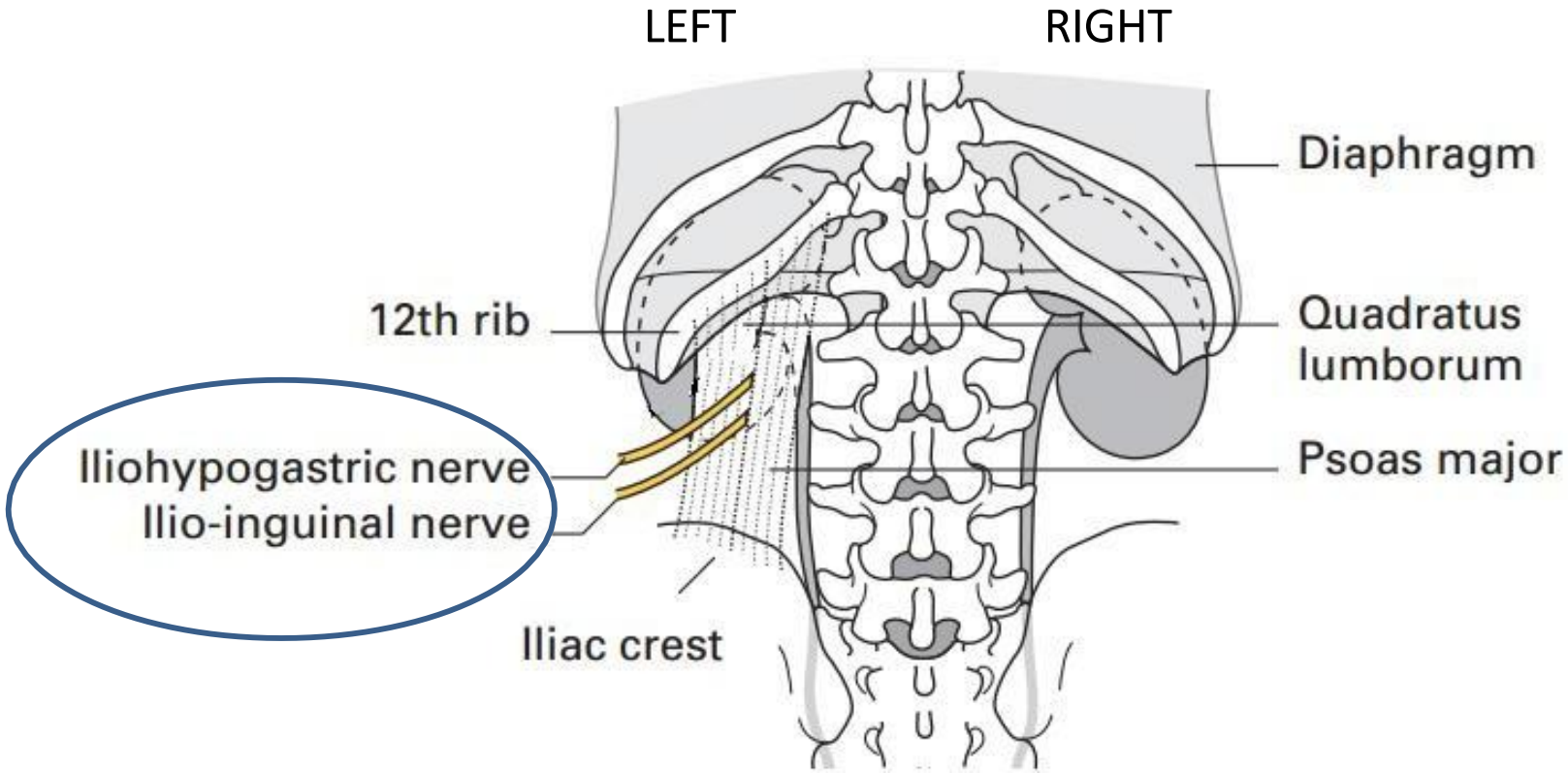
Femoral n.
Obturator n.

Distribution of these 6 nerves




L1
L2
L3
L4
L5
Last Thoracic N.
Ilio-hypogastric
Ilio-inguinal
Genito-femoral
Lat. femoral-cutaneous
Genito-femoral
PSOAS MAJOR
QUADRATUS LUMBORUM
ILIACUS
1st Sacral N.
2nd
3rd
4th
Accessory Obturator
Obturator
Femoral
SARTORIUS
TENSOR FASCIÆ LATÆ
PECTINEUS
ADDUCTOR LONGUS
ADDUCTOR BREVIS
PECTINEUS

Posterior relations of left L1 derived nerves



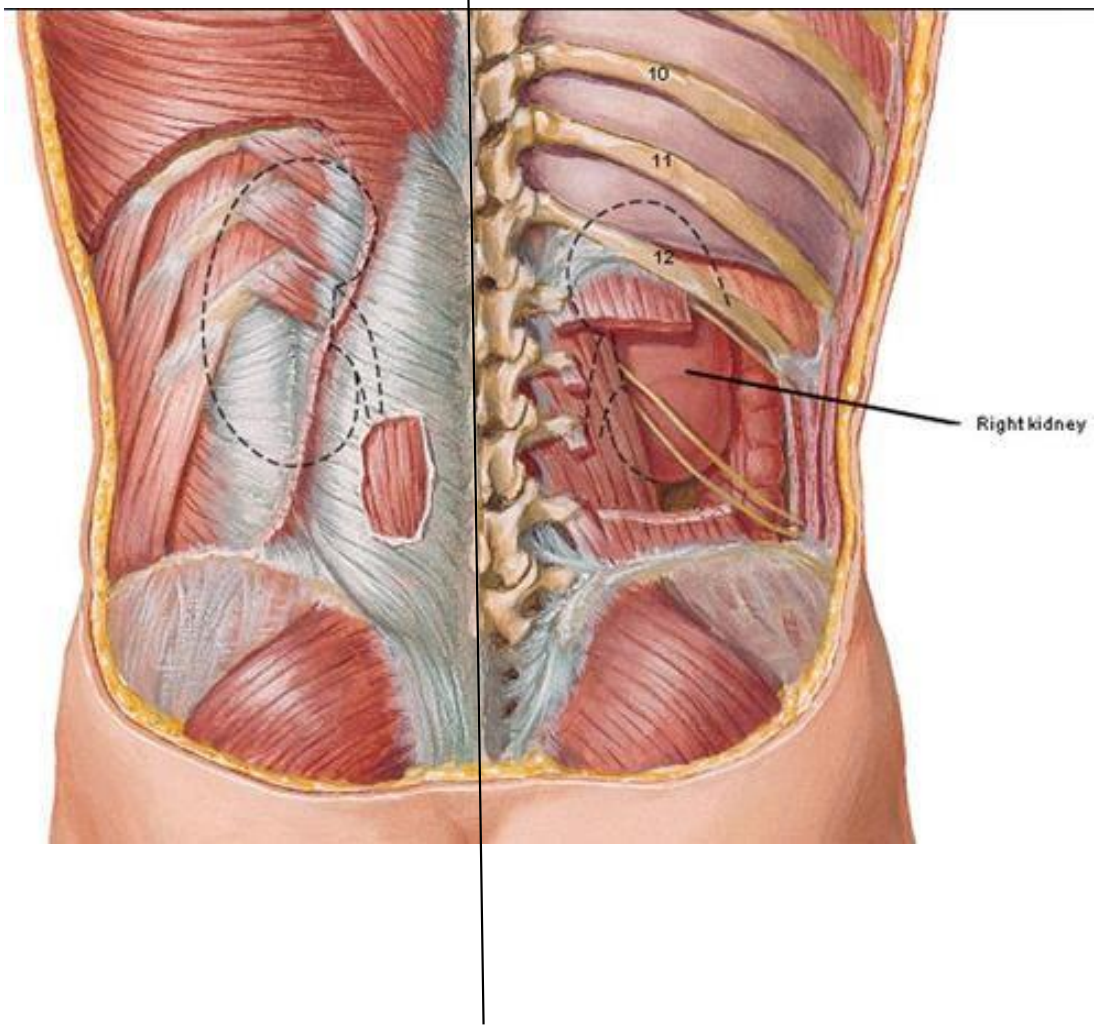
Clinical Anatomy: a Revision and Applied Anatomy for Clinical Students: Written by Harold Ellis,

Posterior dissection of kidney

Superficial

Deep

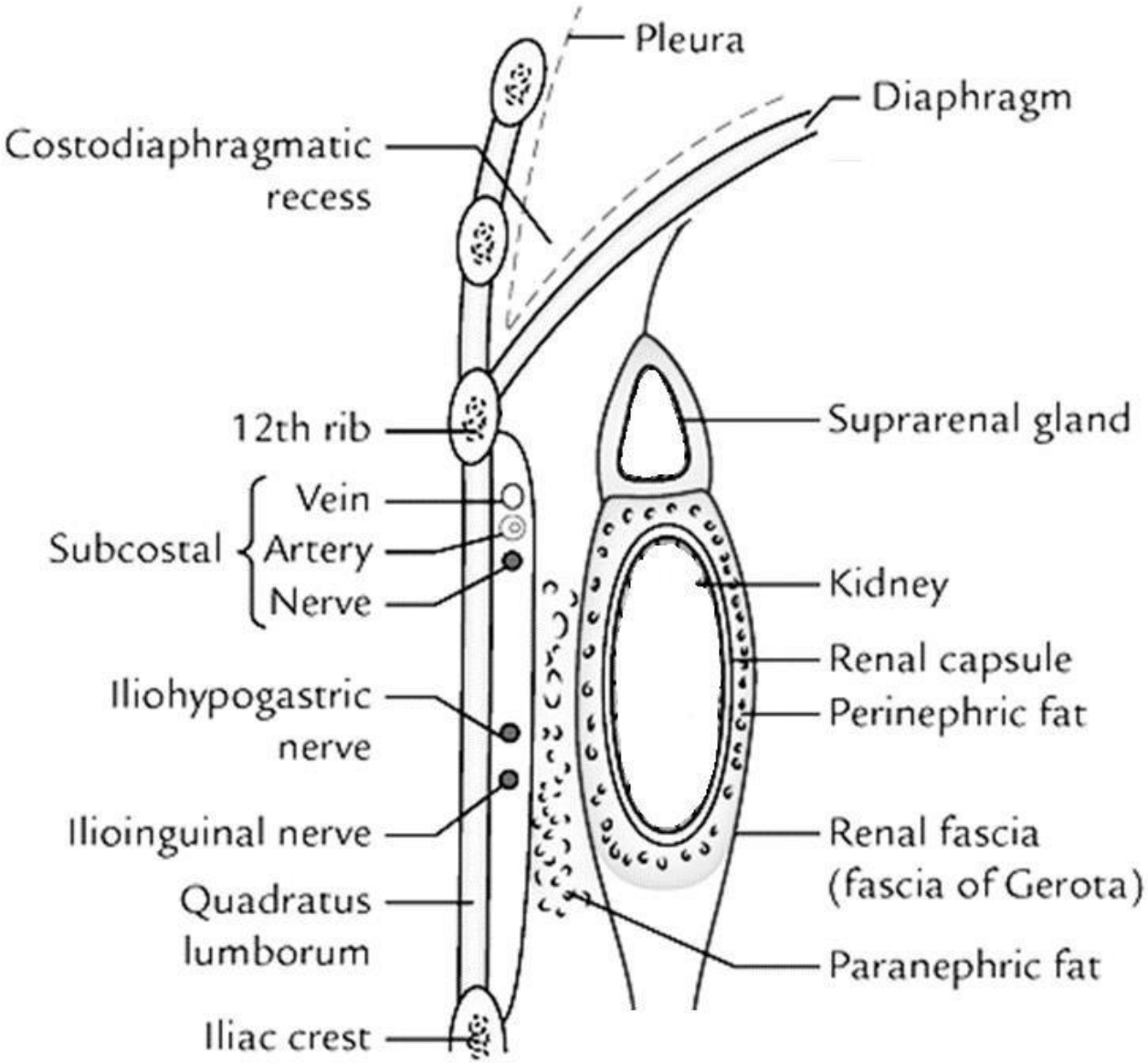
LEFT

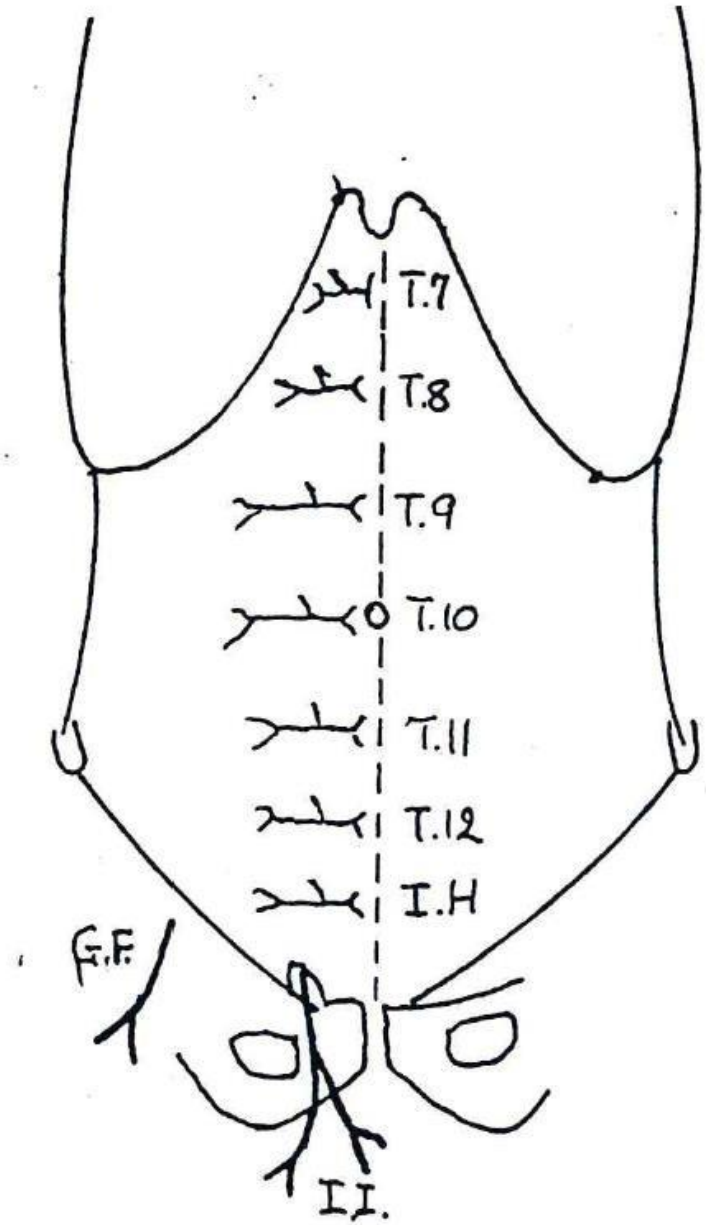


RIGHT

Netter's image

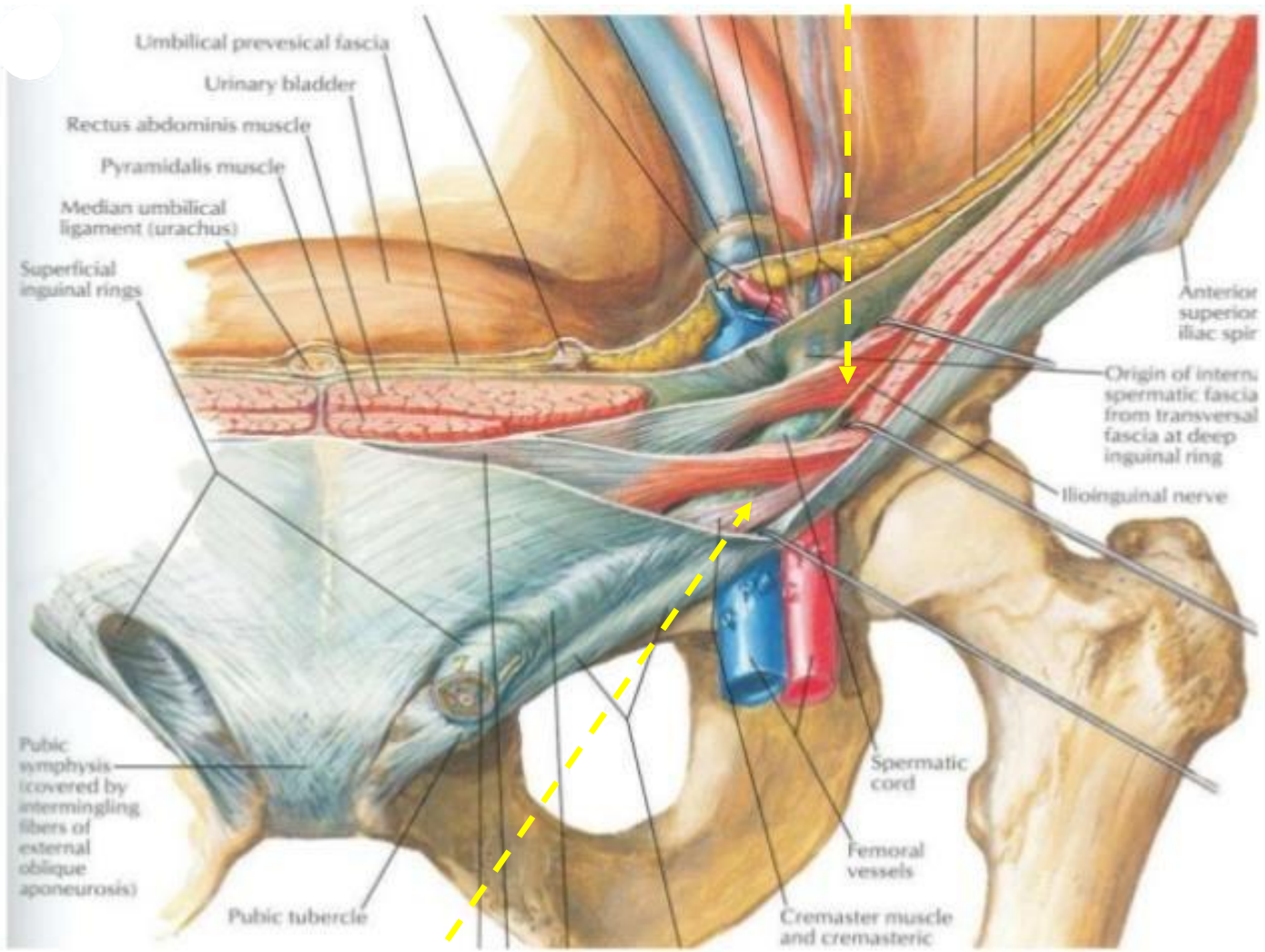
Vertical section through posterior abdominal wall (lumbar region)





IH = iliohypogastric
II = ilioinguinal
GF = genitofemoral

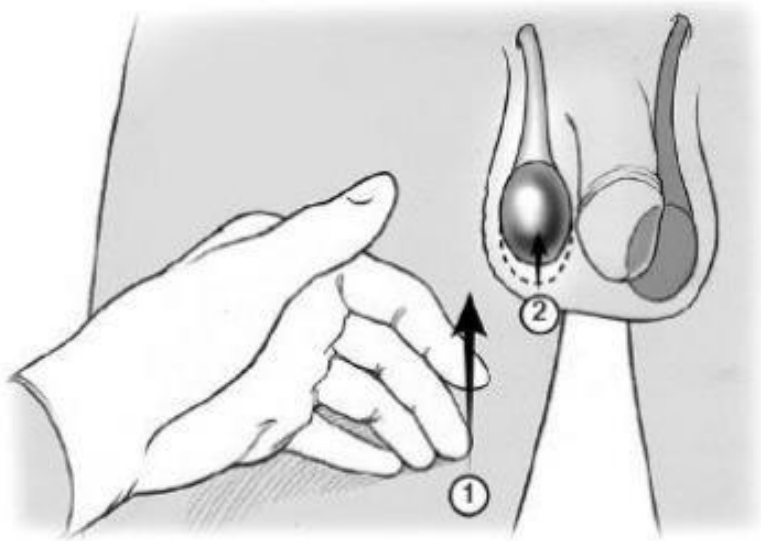
Note the ilioinguinal nerve (in the inguinal canal)



Cremasteric muscle

Netter's image

Cremasteric reflex

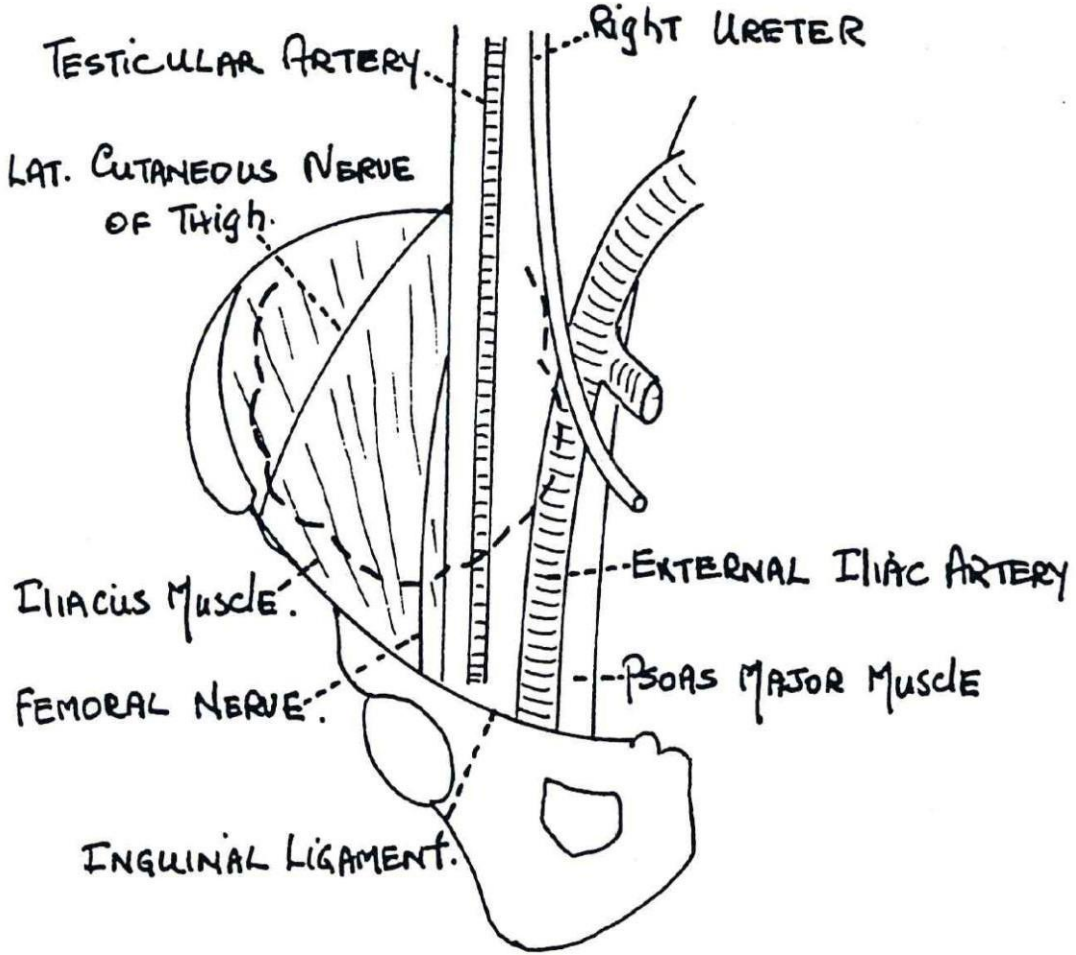


Input is via Ilioinguinal nerve
Output is via Genitofemoral nerve

If absent

- Testicular torsion
- Lower motor neurone disease
- Ilioinguinal nerve injury in hernia repair

Posterior relations of the caecum

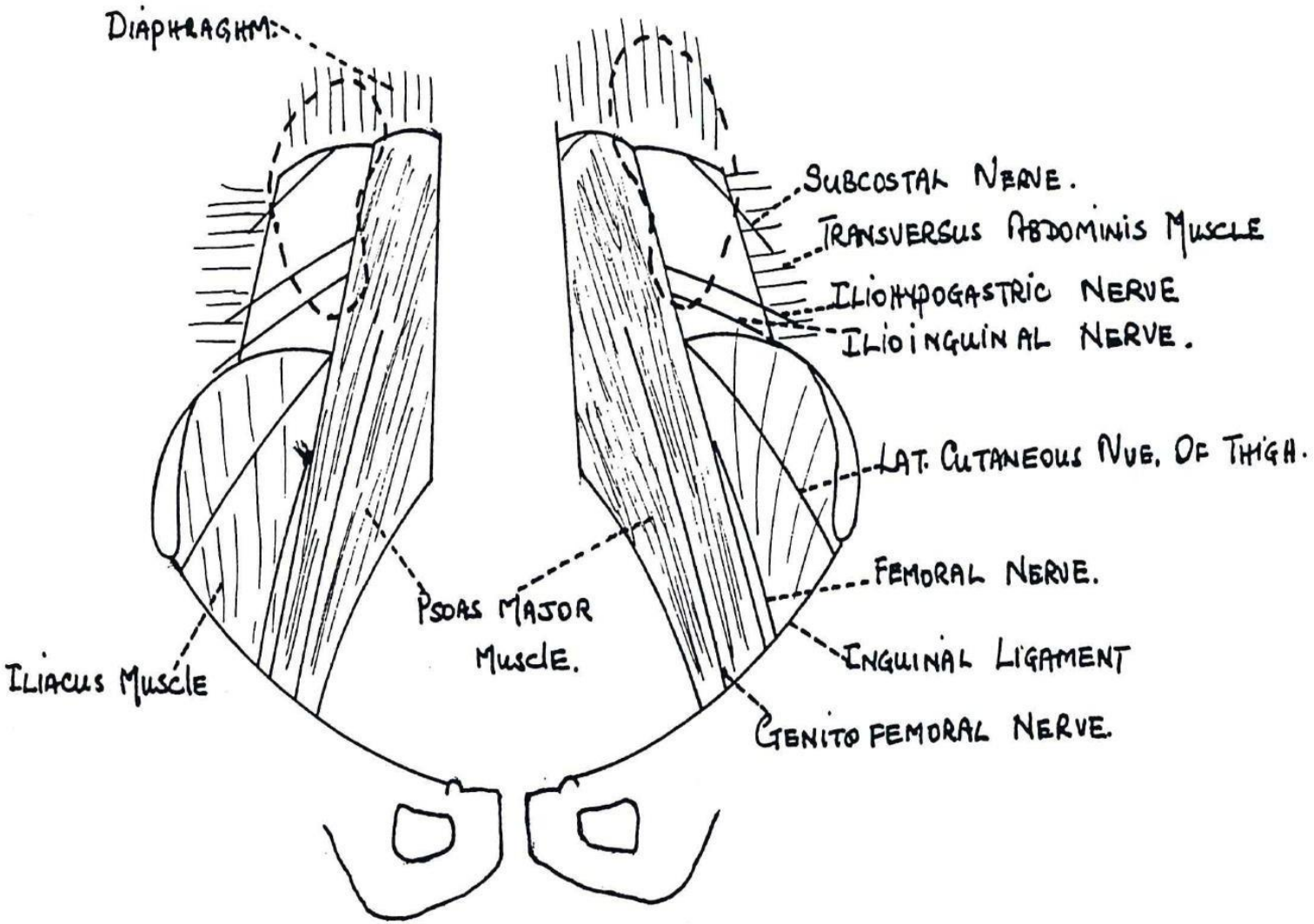


Dissection of posterior abdominal wall with lumbar plexus

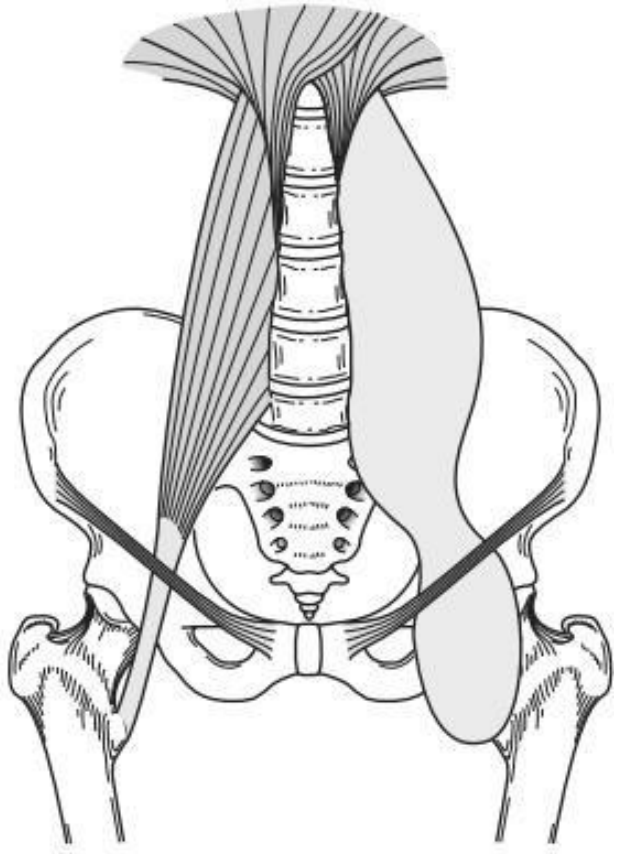


1, psoas major muscle; 2, psoas minor muscle; 3, iliacus muscle; 4, quadratus lumborum muscle; 5, lateral abdominal wall muscles (three layers); 6, kidney; 7, ureter; 8, external iliac artery; 9, sacral promontory; 10, iliohypogastric nerve; 11, ilioinguinal nerve; 12, lateral cutaneous nerve of the thigh; 13, femoral nerve; 14, genitofemoral nerve

Some of the structures forming the posterior abdominal wall

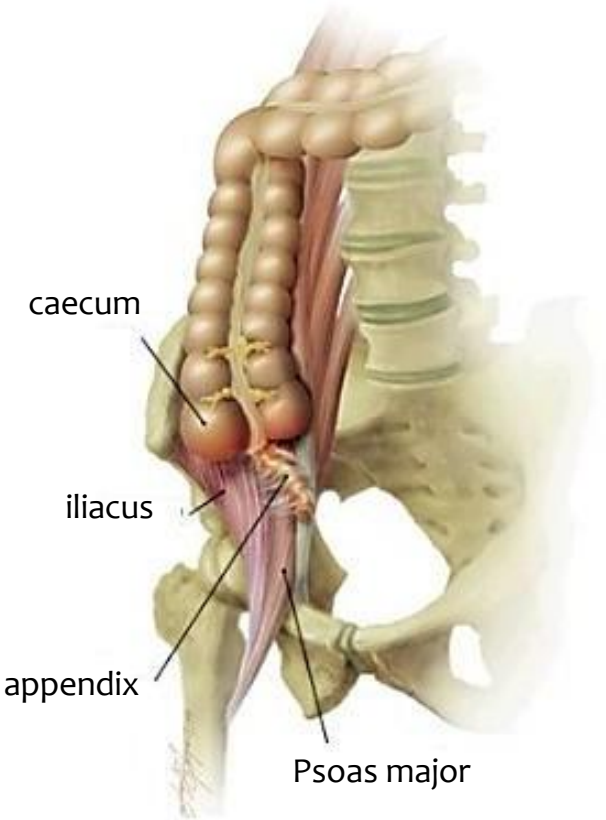


Psoas sheath abscess



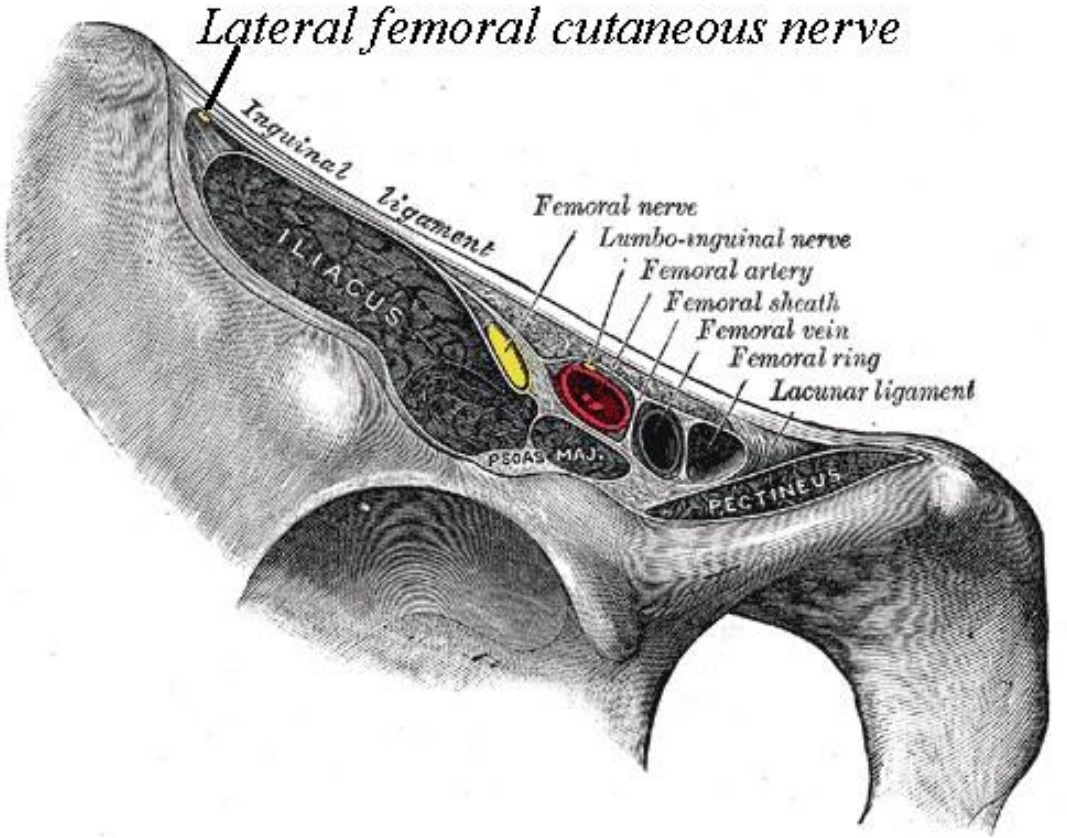
Clinical Anatomy: a Revision and Applied Anatomy for Clinical Students: Written by Harold Ellis,

Psoas sign for acute appendicitis

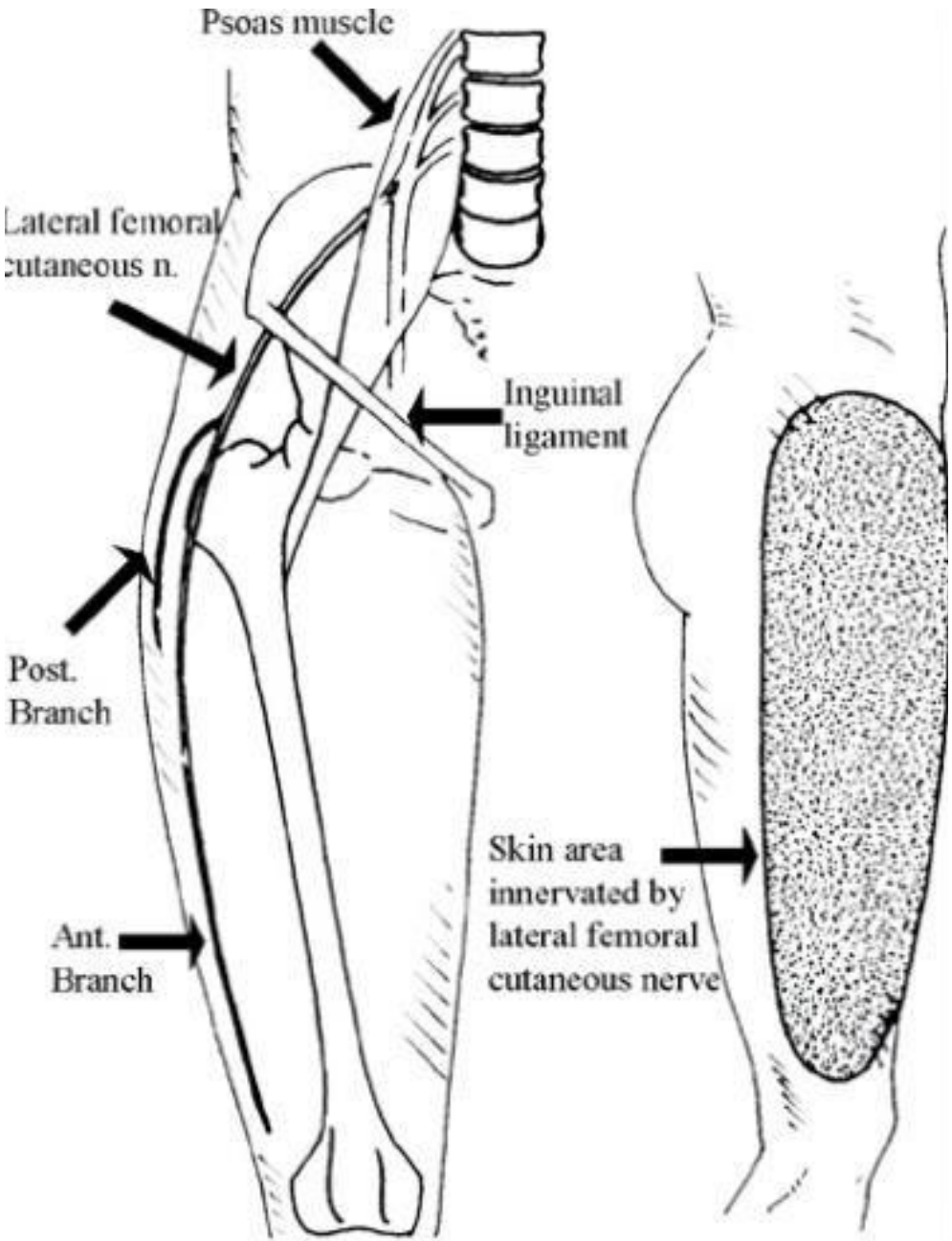




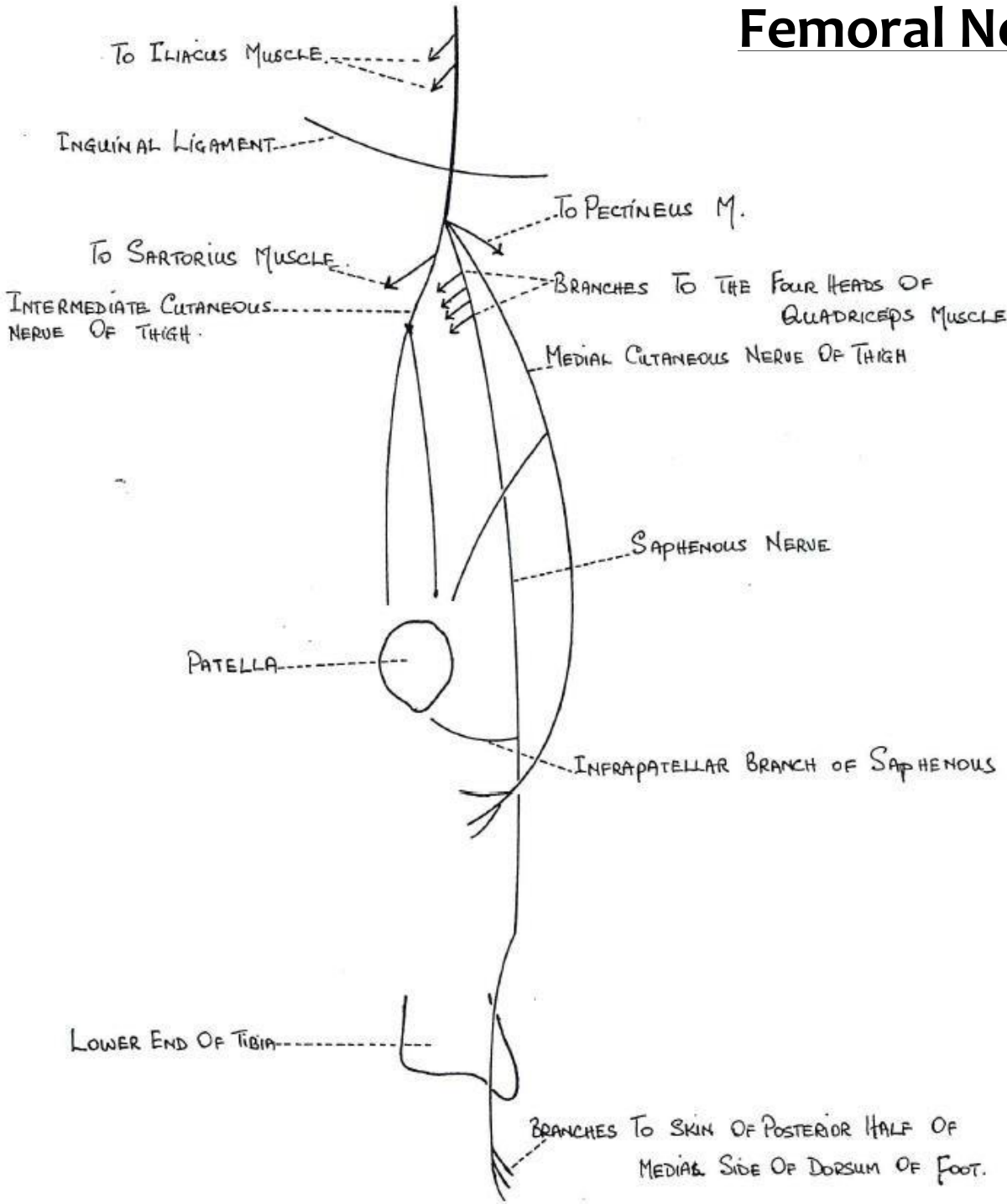
Meralgia paraesthetica



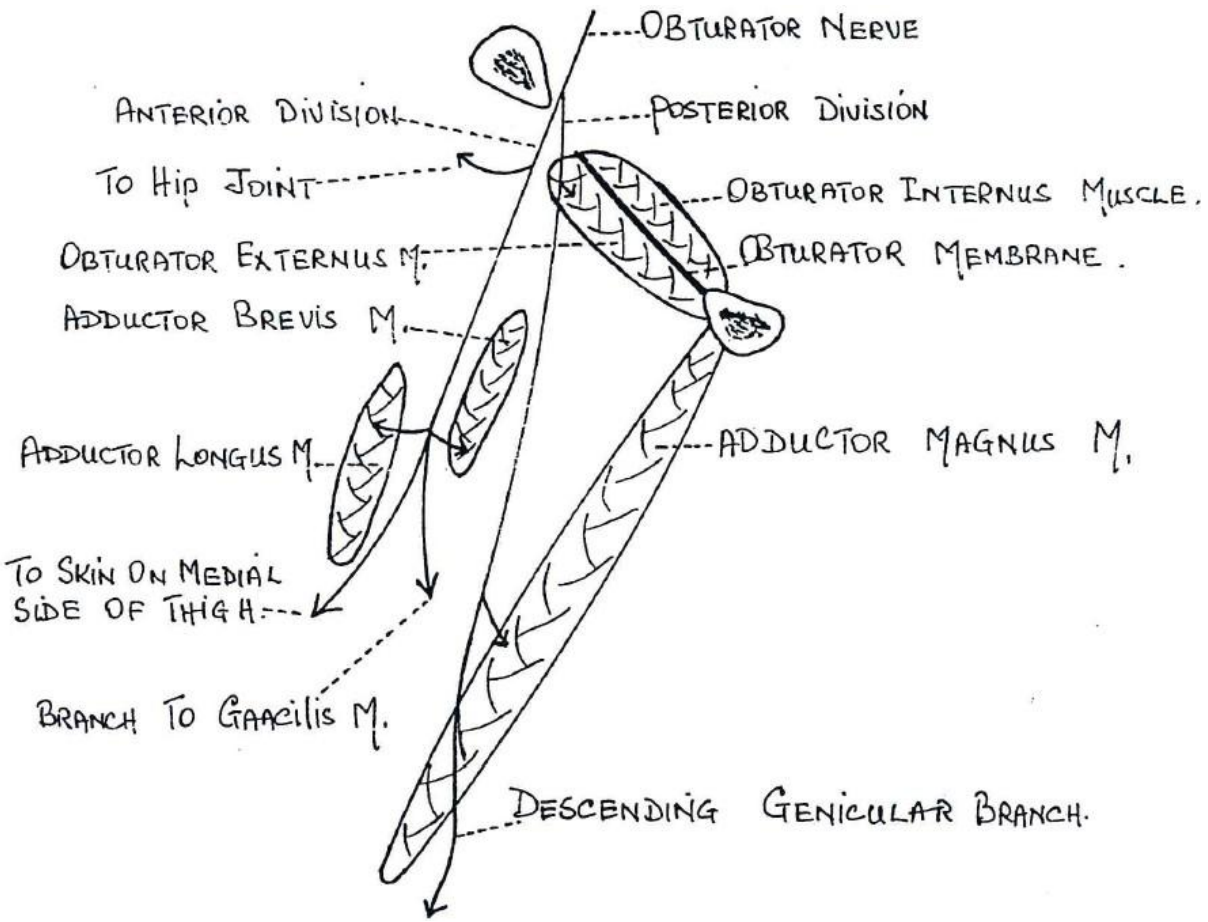
Lateral cutaneous nerve of thigh



Femoral Nerve (L2,3,4)



The distribution of the obturator nerve



Cutaneous innervation of anterior thigh, leg and foot

